Nathan Spencer

Technology Leadership

Contact Information

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Work History

Chief Technology Officer *(Sep. 2024 - Present)*
syGlass, Inc.

- Directly supports school districts, labs, and internal teams in technology integration
- Oversees development of features & products to expand software capabilities in education
- Manages technology integrations, including highly a public intregation with Sony devices
- Sets and revises technical strategy in response to scientific and technological developments
- Plans for reliable scaling of software products based on usage projections
- Leads development team, stablizing roadmap in the face of fast-changing requirements

Lead Software Engineer *(Feb. 2019 - Sep. 2024)*
syGlass, Inc.

- Worked directly with end users to translate functional needs into technical requirements
- Built product roadmaps to make complex research software accessible for classroom use
- Led engineering team through implementation of NIH SBIR Phase II grants worth >$5MM
- Quadrupled software release pace, achieving consistent quarterly cadence
- Offered rapid first-line support to users of the software in both science & education

Software Engineer *(May 2016 - Feb. 2019)*
syGlass, Inc.

- Built hardware-agnostic VR tools and user interfaces using OpenGL, C++
- Worked directly & independently on 3D graphics & transformations, data I/O, UX design
- Collaborated on development tasks monitored via Redmine project management software
- Wrote readable, maintainable code evaluated through peer reviews of pull requests
- Responsible for full project stack: C++ VR app, Python backend, JS & C# UI shells
- Developed through multiple distinct product stages, from premarket to early market

Computer Vision Research Assistant *(Mar. 2015 - Jul. 2015)*
West Virginia University

- Researched & applied machine learning techniques to classify food items in images
- Trained classifiers using deep learning technologies VLFeat and Caffee
- Contributed a highly competitive model tested on 750k images from 101 classes
- Presented results to peers and judges at a WVU research symposium

Technical Skills

- Modern C++
- Git & Github
- OpenGL & GLSL
- Engineering Management
- API Design
- Technical Strategy
- EdTech Integration
- Technical Support
- VR Development
- Python 3
- Image Processing
- Immersive Technology

Education

B.S. Computer Science, Computer Engineering *(Aug. 2014 - May 2018)*
West Virginia University, Morgantown, WV
Dual Degree Program / Magna Cum Laude

Published Works

Godfrey, S.J. *et al.* (2025) 'New light on the trophic ecology of Carcharodon hastalis from teeth embedded in miocene cetacean vertebrae from Calvert Cliffs in Maryland, USA', *Acta Palaeontologica Polonica*, 70(2), pp. 329–337. doi:10.4202/app.01241.2025.